

UNIT 3

LAPLACE TRANSFORM

LM 2 : Laplace Transform of periodic functions

A function $f(x)$ is said to be periodic if and only if $f(x+p)=f(x)$ is true for some value of p and every value of x . The smallest positive value of p for which this equation is true for every value of x will be called the period of the function.

The Laplace Transformation of a periodic function $f(t)$ with period p given by

$$L[f(t)] = \frac{1}{1 - e^{-ps}} \int_0^p e^{-st} f(t) dt$$

Problems

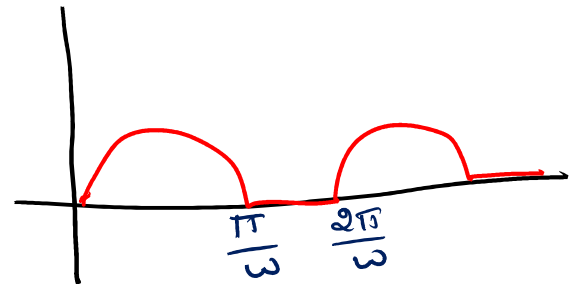
- Find the Laplace transform of the Half wave rectifier function

$$f(t) = \begin{cases} \sin \omega t, & 0 < t < \pi / \omega \\ 0 & \pi / \omega < t < 2\pi / \omega \end{cases}$$

Solution:

This function is a periodic function with period $\frac{2\pi}{\omega}$ in the interval $\left(0, \frac{2\pi}{\omega}\right)$.

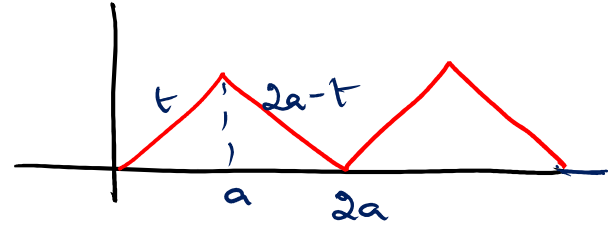
$$\begin{aligned} L[f(t)] &= \frac{1}{1 - e^{-\frac{2\pi s}{\omega}}} \int_0^{\frac{2\pi}{\omega}} e^{-st} f(t) dt \\ &= \frac{1}{1 - e^{-\frac{2\pi s}{\omega}}} \left[\int_0^{\frac{\pi}{\omega}} e^{-st} \sin \omega t dt + 0 \right] \\ &= \frac{1}{1 - e^{-\frac{2\pi s}{\omega}}} \left[\frac{e^{-st}}{s^2 + \omega^2} [-s \sin \omega t - \omega \cos \omega t] \right]_0^{\frac{\pi}{\omega}} \\ &= \frac{1}{1 - e^{-\frac{2\pi s}{\omega}}} \left[\frac{e^{-\frac{\pi s}{\omega}} \omega + \omega}{s^2 + \omega^2} \right] \\ &= \frac{\omega [1 + e^{-\frac{\pi s}{\omega}}]}{[1 - e^{-\frac{\pi s}{\omega}}] [1 + e^{-\frac{\pi s}{\omega}}] s^2 + \omega^2} \\ &= \frac{\omega}{(s^2 + \omega^2) (1 - e^{-\frac{\pi s}{\omega}})} \end{aligned}$$



2. Find the Laplace transform of a triangular wave is given by $f(t) = \begin{cases} t, & 0 < t < a \\ 2a - t & a < t < 2a \end{cases}$ with $f(t+2a) = f(t)$.

Solution:

$$\begin{aligned}
 L[f(t)] &= \frac{1}{1-e^{-2as}} \int_0^{2a} e^{-st} f(t) dt \\
 &= \frac{1}{1-e^{-2as}} \left[\int_0^a e^{-st} t dt + \int_a^{2a} e^{-st} (2a-t) dt \right] \\
 &= \frac{1}{1-e^{-2as}} \left[t \left(\frac{e^{-st}}{-s} \right) - (1) \left(\frac{e^{-st}}{s^2} \right) \right]_0^a + \left[(2a-t) \left(\frac{e^{-st}}{-s} \right) - (-1) \left(\frac{e^{-st}}{s^2} \right) \right]_a^{2a} \\
 &= \frac{1}{1-e^{-2as}} \left[\left[-t \frac{e^{-st}}{s} - \frac{e^{-st}}{s^2} \right]_0^a + \left[-(2a-t) \frac{e^{-st}}{s} + \frac{e^{-st}}{s^2} \right]_a^{2a} \right] \\
 &= \frac{1}{1-e^{-2as}} \left[\left(-a \frac{e^{-sa}}{s} - \frac{e^{-sa}}{s^2} \right) - \left(-\frac{1}{s^2} \right) \right] + \left[\left(\frac{e^{-2as}}{s^2} \right) - \left(-\frac{ae^{-as}}{s} + \frac{e^{-as}}{s^2} \right) \right] \\
 &= \frac{1}{1-e^{-2as}} \left[\frac{-ae^{-sa}}{s} - \frac{e^{-as}}{s^2} + \frac{1}{s^2} + \frac{e^{-2as}}{s^2} + \frac{ae^{-as}}{s} - \frac{e^{-as}}{s^2} \right] \\
 &= \frac{1}{1-e^{-2as}} \left[\frac{1+e^{-2as}-2e^{-as}}{s^2} \right] \\
 &= \frac{[1-e^{-as}]^2}{s^2(1-e^{-as})(1+e^{-as})} \\
 &= \frac{[1-e^{-as}]}{s^2(1+e^{-as})} \\
 &= \frac{1}{s^2} \tanh \left[\frac{as}{2} \right]
 \end{aligned}$$

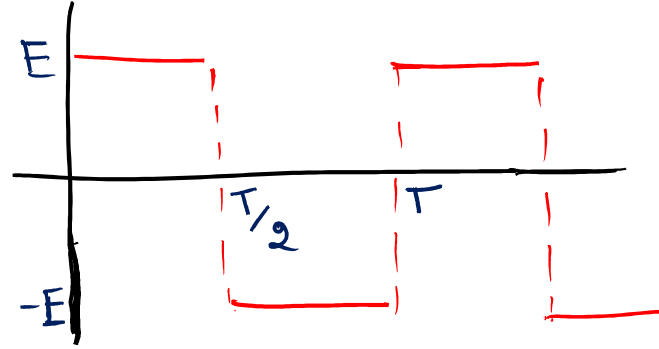


3. Find the Laplace transform of a triangular wave is given by $f(t) = \begin{cases} E, & 0 < t < \frac{T}{2} \\ -E, & \frac{T}{2} < t < T \end{cases}$

with $f(t+T) = f(t)$

Solution:

$$\begin{aligned}
 L[f(t)] &= \frac{1}{1-e^{-Ts}} \int_0^T e^{-st} f(t) dt \\
 &= \frac{1}{1-e^{-Ts}} \left[\int_0^{\frac{T}{2}} e^{-st} E dt + \int_{\frac{T}{2}}^T e^{-st} (-E) dt \right] \\
 &= \frac{E}{1-e^{-Ts}} \left[\int_0^{\frac{T}{2}} e^{-st} dt - \int_{\frac{T}{2}}^T e^{-st} dt \right] \\
 &= \frac{E}{1-e^{-Ts}} \left[\left(\frac{e^{-st}}{-s} \right) \right]_0^{\frac{T}{2}} - \left[\left(\frac{e^{-st}}{-s} \right) \right]_{\frac{T}{2}}^T \\
 &= \frac{-E}{s(1-e^{-Ts})} \left[\left(e^{-\frac{sT}{2}} - 1 \right) - \left(e^{-sT} - e^{-\frac{sT}{2}} \right) \right] \\
 &= \frac{-E}{s(1-e^{-Ts})} \left[-1 - e^{-sT} + 2e^{-\frac{sT}{2}} \right] \\
 &= \frac{E}{s(1-e^{-Ts})} \left[\left(e^{-\frac{sT}{2}} - 1 \right)^2 \right] \\
 &= \frac{E}{s} \frac{\left[1 - e^{-\frac{sT}{2}} \right]^2}{\left(1 - e^{-\frac{sT}{2}} \right) \left(1 + e^{-\frac{sT}{2}} \right)} \\
 &= \frac{E}{s} \frac{\left(1 - e^{-\frac{sT}{2}} \right)}{\left(1 + e^{-\frac{sT}{2}} \right)} \\
 &= \frac{E}{s} \tanh \left[\frac{sT}{4} \right]
 \end{aligned}$$



UNIT – 3

INVERSE LAPLACE TRANSFORMS

LM1: Inverse Laplace Transforms

If the Laplace transform of the time domain function $f(t)$ is

$$F(s) = L[f(t)] = \int_0^{\infty} e^{-st} f(t) dt$$

then the inverse Laplace transform $F(s)$ is the time domain function $f(t)$. That is,

$$L^{-1}[F(s)] = f(t)$$

LT of some elementary functions

$f(t)$	$L[f(t)]$
1	$\frac{1}{s}$
e^{at}	$\frac{1}{s-a}$
e^{-at}	$\frac{1}{s+a}$
$\cos at$	$\frac{s}{s^2 + a^2}$
$\sin at$	$\frac{a}{s^2 + a^2}$
$\cosh at$	$\frac{s}{s^2 - a^2}$
$\sinh at$	$\frac{a}{s^2 - a^2}$
t^n	$\frac{n!}{s^{n+1}}$

ILT of some elementary functions

$F(s)$	$L^{-1}[F(s)]$
$\frac{1}{s}$	1
$\frac{1}{s-a}$	e^{at}
$\frac{1}{s+a}$	e^{-at}
$\frac{s}{s^2+a^2}$	$\cos at$
$\frac{a}{s^2+a^2}$	$\sin at$
$\frac{s}{s^2-a^2}$	$\cosh at$
$\frac{a}{s^2-a^2}$	$\sinh at$
$\frac{n!}{s^{n+1}}$	t^n

Examples:

$$1. \quad L^{-1}\left[\frac{1}{s^2+1}\right] = \sin t$$

$$2. \quad L^{-1}\left[\frac{s}{s^2+9}\right] = L^{-1}\left[\frac{s}{s^2+3^2}\right] = \cos 3t$$

$$3. \quad L^{-1}\left[\frac{6}{s}\right] = 6L^{-1}\left[\frac{1}{s}\right] = 6(1) = 6$$

$$4. \quad L^{-1}\left[\frac{2}{s+5}\right] = 2L^{-1}\left[\frac{1}{s+5}\right] = 2e^{-5t}$$

$$5. \quad L^{-1}\left[\frac{4}{s-3}\right] = 4L^{-1}\left[\frac{1}{s-3}\right] = 4e^{3t}$$

$$6. \quad L^{-1}\left[\frac{2}{t^3}\right] = L^{-1}\left[\frac{2!}{t^{2+1}}\right] = t^2$$

$$7. \quad L^{-1}\left[\frac{1}{t^4}\right] = L^{-1}\left[\frac{1}{t^{3+1}}\right] = \frac{1}{3!}L^{-1}\left[\frac{3!}{t^{3+1}}\right] = \frac{1}{6}(t^3) = \frac{t^3}{6}$$

Convolution:

The convolution of the functions $f(t)$ & $g(t)$ is defined as

$$f * g = \int_0^t f(u)g(t-u)du$$

Convolution Theorem:

$$L^{-1}[\phi_1(s)\phi_2(s)] = L^{-1}[\phi_1(s)] * L^{-1}[\phi_2(s)]$$

Problems:

1. Find $L^{-1}\left[\frac{1}{(s+3)(s-1)}\right]$

Solution:

By convolution Theorem,

$$L^{-1}\left[\frac{1}{(s+3)(s-1)}\right] = L^{-1}\left[\frac{1}{(s+3)} \frac{1}{(s-1)}\right] = L^{-1}\left[\frac{1}{(s+3)}\right] * L^{-1}\left[\frac{1}{(s-1)}\right] = e^{-3t} * e^t$$

Now,

$$e^{-3t} * e^t = \int_0^t e^{-3u} e^{t-u} du = e^t \int_0^t e^{-4u} du = e^t \left[\frac{e^{-4u}}{-4} \right]_0^t = \frac{1}{4} [e^t - e^{-3t}]$$

2. Find $L^{-1}\left[\frac{1}{(s-2)(s+2)^2}\right]$

Solution:

By convolution Theorem,

$$L^{-1}\left[\frac{1}{(s-2)(s+2)^2}\right] = L^{-1}\left[\frac{1}{(s-2)} - \frac{1}{(s+2)^2}\right] = L^{-1}\left[\frac{1}{(s-2)}\right] * L^{-1}\left[\frac{1}{(s+2)^2}\right] = e^{2t} * te^{-2t}$$

Now,

$$e^{2t} * te^{-2t} = \int_0^t e^{2u}(t-u)e^{-2(t-u)} du = e^{-2t} \int_0^t (t-u)e^{4u} du$$

$$= e^{-2t} \left[(t-u) \frac{e^{4u}}{4} + \frac{e^{4u}}{16} \right]_0^t$$

$$= e^{-2t} \left[\left(0 + \frac{e^{4t}}{16} \right) - \left(t \frac{e^{4t}}{4} + \frac{1}{16} \right) \right]$$

$$= \frac{e^{2t}}{16} - t \frac{e^{2t}}{4} + \frac{e^{-2t}}{16}$$

Practice Problems:

1. Find $L^{-1}\left[\frac{5}{(s+2)^5}\right]$

2. Find $L^{-1}\left[\frac{1}{s(s+1)^2}\right]$